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1 Introduction

This project is composed of two parts. In the first part, we will implement a numerical model of a lower mass galaxy fly-by next to a higher mass galaxy. In the second part, we will model the collapse of a star. Both models use n-body simulations.

2 Implementation

3 Results

3.1 Galaxy Merger

3.2 Stellar Collapse

4 Discussion

- Why did you *need* to perform a numerical simulation to solve this astrophysical problem?
- Motivate the validity of the models used in this project. What physical aspects are well captured by them? What are their limitations?
- What physical ingredients could be improved / added in order to make the models more *realistic*?